Statistical and Economic Evaluation of Volatility Term-Structure Forecasts: A Paired Placebo Design Applied to VIX Futures

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Abstract

When a forecasting model is evaluated, the benchmark is the random walk. When a *strategy* built on that model is evaluated, it rarely is. We run the nested null through the entire economic pipeline, keeping the same threshold, risk targeting, costs and days, so that two portfolios differ in the forecast and in nothing else, and the overlap in their positions bounds what the forecast contributes. We use this to ask whether the S&P 500 implied-volatility term structure can be forecast, and whether the forecast can be monetized on VIX futures.

Over 2010–2023, with an eight-year out-of-sample window, it contributes nothing we can measure. The strategy shows a Sharpe ratio of 0.29 to 0.35 and the placebo 0.44 to 0.46 on the same days; the paired difference runs from -0.09 to -0.17, and eight years bound the contribution above by $+0.40$ of Sharpe. Across forty-five configurations of threshold, risk target and horizon, a test of superior predictive ability returns $p = 0.82$.

Three refinements of the inference point the same way. A block-bootstrap null for the multi-step Clark–West statistic is right-skewed at long horizons, and the monthly statistics cease to be significant. A Romano–Wolf stepdown, exploiting the dependence across the forty-two tests, retains seventeen rather than Holm's seven, but no monthly forecast of the thirty-day point. And the spanning regression in traded space is a mechanical artefact of the futures price's persistence: an information-free forecast reproduces it, and freeing the coefficient restriction leaves the forecast with $t = 0.10$.

What remains is a separation between the object forecast and the object traded: the VIX is a variance-swap rate while our target is an at-the-money volatility, and the constant-maturity price is not what a rolling position earns. Modelling the futures curve directly recovers no signal. What the position earns is the volatility risk premium.

Keywords: forecast evaluation, nested benchmarks, implied volatility, term structure, Nelson–Siegel, VIX, volatility risk premium, multiple testing, data snooping.
JEL classification: G13, G14, C22, C53.

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1 Introduction

An option's price embeds a forecast. Inverting the Black–Scholes formula on a traded option returns the implied volatility, the volatility that reconciles model and market, and reading it across maturities produces a curve: how much movement the market expects at one month, three months, a year. That curve is the object of this paper. Its thirty-day point is the CBOE VIX index, which, unlike the curve as a whole, can be traded directly through futures.

The curve moves in an orderly way. It usually slopes upward, because short-dated volatility is low in quiet markets and mean-reverts upward; in a crisis it inverts, because the immediate risk exceeds anything expected a year out. That regularity invites an obvious question. If the shape is informative, can it be used to forecast where volatility is going, and can that forecast be monetized? Guo, Han and Zhao (2014) answer the first half affirmatively, modelling the curve with a Nelson–Siegel decomposition and documenting factor predictability. We take the question to its conclusion.

Related work. The pattern we document, statistical predictability without economic value, is not new and we do not claim it. Gonçalves and Guidolin (2006) find predictability in the S&P 500 implied-volatility surface that does not survive trading costs, and Konstantinidi, Skiadopoulos and Tzagkaraki (2008) reach the same verdict for implied volatility indices, testing economic significance on CBOE volatility futures much as we do. What is new is the source we identify: up to a week the curve's own forward volatility forecasts at least as well as anything we build, and beyond a week what looks like identifiable predictive content for the futures price turns out to be generated by the persistence of that price rather than by the forecast. That connects the result to the term-structure literature, where Johnson (2017) shows the slope of the VIX term structure pricing variance risk premia rather than expected VIX changes, and Simon and Campasano (2014) find the futures basis failing to forecast the VIX while a basis-referenced short-volatility trade is profitable. We agree with both, and Section 5 makes the correspondence explicit.

The benchmark we propose. A methodological point runs through the paper. Forecast evaluation has a settled discipline: the competitor is the random walk, it is nested in every model, and tests exist that respect that nesting. Economic evaluation has only a partial convention. Since Campbell and Thompson (2008) the return-predictability literature has compared a model-driven allocation against the same allocation driven by the historical mean, a practice

consolidated by Rapach, Strauss and Zhou (2010). What is less often done, and what we do here, is to push the nested null through *every* stage of the pipeline, keeping the same threshold, risk targeting, projection, costs and days, so that the two portfolios differ in the forecast and in nothing else. Because they then hold overlapping positions, the paired difference in their Sharpe ratios has a standard error smaller than either, which is what lets an eight-year sample bound the contribution rather than merely fail to detect it. Inference on such a difference is the problem posed by Jobson and Korkie (1981) and made robust by Ledoit and Wolf (2008). The gain is proportional to the correlation between the two arms, however, and Section 5 shows it disappearing when that correlation turns negative.

Three questions, in order. Each section answers one before the next is posed. Can we forecast the curve out-of-sample, and does the answer depend on the model (Section 3)? Is the forecast distinct from what the futures curve already prices, and what rule turns it into a position (Section 4)? Does the strategy earn anything net of costs that the same rule without a forecast does not (Section 5)?

The short answers are that forecastability exists but is weak, belongs to no model and does not survive a bootstrap reference at the monthly horizon, that the market's own quote is at least as good out to a week, and that the trading rule earns the volatility risk premium and nothing else. We state this narrowly. It is not a test of market efficiency, which a single crude rule on a single instrument cannot deliver: what we establish is that *relative to a nested benchmark that uses no forecast*, the forecast delivers no measurable incremental economic value. The claim is obtained on an independent fourteen-year sample with inference designed to resist the usual sources of false discovery.

2 The volatility curve and its three factors

2.1 Data and construction

The main sample is the free end-of-day S&P 500 option-chain archive distributed by optionsDX (`https://www.optionsdx.com`), covering 2010–2023: one file per month, each row a contract with its quote timestamp, expiry, strike, bid, ask and the underlying index level. We use the raw quotes only, and every implied volatility in the main text is inverted by us. For each day and expiry we first recover the forward and the discount factor from the option prices themselves, using the fact that the call-minus-put price is affine in the strike,

$$C(K) - P(K) = e^{-rT}\,(F - K), \tag{1}$$

so that a cross-sectional regression of $C(K) - P(K)$ on K returns the discount factor as minus the slope and the forward as the intercept divided by it, with no assumption about rates or dividends. We then take the out-of-the-money side at the strike nearest that forward and invert Black–Scholes numerically on the mid-quote.

One consequence of recovering the forward this way deserves to be stated, because it removes an objection rather than raising one. VIX futures settle at 16:15 ET, fifteen minutes after our option snapshot, so a construction that measured moneyness against the index level printed in the file would inherit that asynchrony. Ours does not: the forward comes from the option prices themselves through Equation (1), and the index level in the file is never used.

Quotes are kept if the bid is strictly positive, the ask no lower than the bid, and the relative spread below 25% of the mid. That last filter is permissive for at-the-money SPX options, whose spread is usually well under five percent, and it is deliberately so: on an end-of-day file a tight filter discards whole expiries at the long end, where the fit needs anchoring. The cost is that some noise enters, and Section 3.4 shows where it goes. A day is retained only if it offers at least four distinct expiries, the shortest at twenty-one days or less and the longest at ninety or

more; expiries under five days are excluded from the fit, since event and week-end premia distort them. The archive then yields 3,354 usable trading days with a median of 32 maturities per day, one snapshot each at the 16:00 close. A second construction, which reads the vendor's own implied volatilities at the listed strike nearest spot, serves as a robustness check: the two curves correlate 0.997 with a mean absolute difference of 0.45 volatility points and give the same answer everywhere we have compared them (Appendix B).

The option sample stops at the end of 2023 because that is where the vendor's free release ended; the cut-off is a constraint, not a choice, and no year was dropped after inspecting results. Our baseline convention assumes a signal read at the 16:00 snapshot is executed at the 16:15 futures settlement. That is realistic but not free, so Section 5 also reports the conservative alternative in which a full trading day separates signal from execution. *Out-of-sample* means 2016–2023 throughout, with 2010–2015 used for training and identification. Our design choices were made knowing the full sample; for a negative result that latitude works against the conclusion, since specification search biases an evaluation towards finding profitability and we find none.

One auxiliary dataset appears later. CBOE VIX futures settlements (ticker VX, 163 contracts) run from January 2013 to June 2026 and provide the tradable instrument of Section 5; the constant-maturity series built from them ends in February 2026, since the last months of the record do not yet bracket every maturity we need. Every result that needs an option curve is therefore confined to 2016–2023, while the few that need only futures use the full window, which we say where it matters, because the two samples give different answers and that difference is itself informative.

2.2 Three numbers instead of thirty

Tracking thirty maturities a day is unwieldy and mostly redundant, since the curve moves in a few coordinated ways. Following Nelson and Siegel (1987), and Guo, Han and Zhao (2014) for the volatility application, we summarize it each day by three factors:

$$\mathrm{IV}(\tau) = \underbrace{\beta_0}_{\text{level}} + \beta_1 \underbrace{\frac{1 - e^{-\tau/\lambda}}{\tau/\lambda}}_{\text{decays from 1 to 0}} + \beta_2 \underbrace{\left(\frac{1 - e^{-\tau/\lambda}}{\tau/\lambda} - e^{-\tau/\lambda} \right)}_{\text{hump, peaks in the middle}}. \tag{2}$$

The level β_0 is the long-maturity asymptote. The slope β_1 loads on short maturities and dies away, so it measures how far the front of the curve sits below (contango) or above (backwardation) the long end. The curvature β_2 has a loading that is zero at both ends and peaks in the middle. Factors are estimated daily by ridge-regularized least squares, the penalty applying to slope and curvature only so as to stabilize the shape without biasing the height. Treating the loadings as fixed and letting the factors move is the dynamic reading of Nelson–Siegel introduced by Diebold and Li (2006) for bond yields.

Guo, Han and Zhao write the loadings as $e^{-\lambda\tau}$ with τ in days and report $\lambda = 0.0147\,\mathrm{day}^{-1}$; Equation (2) uses the reciprocal convention, in which λ is a time constant equal to $1/0.0147 \approx 68$ days, or 0.186 years. A grid search on our data locates the same optimum, but it uses the full sample and is therefore in-sample, so we adopt the published value and treat the agreement as corroboration rather than calibration. Estimating λ recursively alongside the three factors would require non-linear least squares every day, which on volatility surfaces is unstable and prone to local optima; fixing it, as Diebold and Li (2006) do for bond yields, keeps the factors comparable across time, and whatever sub-optimality that costs works against finding predictability rather than for it. The median cross-sectional R^2 is 0.966 over fourteen years; Figure 1 shows the fit in both regimes and Table 1 summarizes the resulting series and factors.

Volatility rises with maturity on average while its dispersion falls, and persistence increases with maturity. The extremes, such as the 95% nine-day implied volatility of March 2020, are the

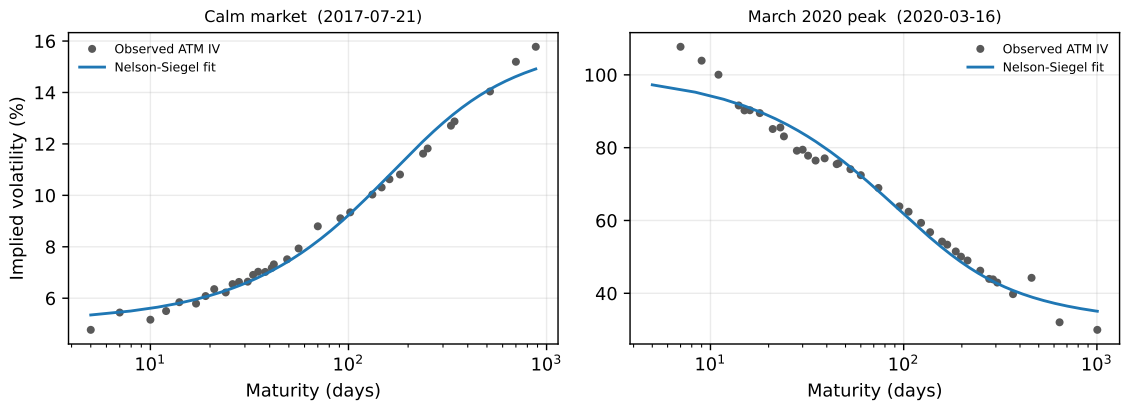


Figure 1: The curve on a calm day (left) and at the peak of the March 2020 crisis (right). Dots are observed at-the-money implied volatilities, the line is the three-factor fit; the maturity axis is logarithmic. Note the inversion: in calm markets the curve rises from about 5% at a week to 16% at the long end, while on 16 March 2020 it falls from 108% at a week to 30% almost three years out.

Table 1: Descriptive statistics, 2010–2023. Implied volatilities (%) are reconstructed from the daily three-factor fit. This does not inflate persistence: the raw interpolated IV_{30} has autocorrelations of 0.960 and 0.612 at one and twenty-one days against 0.956 and 0.582 for the fitted series, and the two correlate at 0.996. Appendix A repeats the forecast evaluation on the raw target.

	Mean	Std	Min	Max	AC1	AC21
IV_9	15.28	7.56	5.17	94.85	0.941	0.507
IV_{30}	15.69	6.81	6.34	83.93	0.956	0.582
IV_{60}	16.22	6.07	7.53	72.30	0.970	0.662
IV_{90}	16.70	5.56	8.54	64.00	0.977	0.714
IV_{180}	17.74	4.62	10.78	53.17	0.984	0.786
β_0 (level)	20.14	3.03	15.05	36.51	0.971	0.827
β_1 (slope)	−5.05	6.28	−14.22	68.63	0.926	0.405
β_2 (curvature)	−2.26	6.29	−95.21	25.09	0.820	0.407

crisis inversions. ADF rejects a unit root for all four series while KPSS rejects stationarity, the joint signature of a highly persistent but mean-reverting process, so no differencing is required and ARIMA collapses to ARMA.

3 Can we forecast the curve?

3.1 Setup

We forecast the three factors and the implied volatility itself at horizons of $1, 2, 5, 7, 14$ and 30 trading days. The benchmark is the **random walk**, "tomorrow equals today", which on a series this persistent is demanding. Three pure time-series competitors follow: an **ARMA** on the series alone, with orders chosen by BIC on the training window and then held fixed; the **HAR** of Corsi (2009), combining daily, weekly and monthly averages; and a **VAR(2)** on the three factors jointly, recomposed into a volatility. The fourth, a **ridge** regression, is the only one that conditions on the shape of the curve, predicting the change in volatility from four features: the level IV_{30}, its five-day change, the long slope $\mathrm{IV}_{90} - \mathrm{IV}_{30}$ and the short slope $\mathrm{IV}_{30} - \mathrm{IV}_9$. Its penalty is fixed a priori at $\alpha = 15$ and never tuned; over $\alpha \in \{1, \ldots, 150\}$ the out-of-sample skill moves by at most 1.4 percentage points, and by 0.4 at thirty days. The daily Nelson–Siegel fit

uses a separate and much smaller ridge, $\alpha_{\mathrm{NS}} = 0.01$, on the slope and curvature loadings only.

Everything is estimated walk-forward. The time-series models are re-estimated every 126 trading days and updated by filtering in between; the ridge, a closed form refitted at every date, is exactly reproducible. Models with explanatory variables additionally embargo the last h observations of the training window, so that no training target overlaps the date being forecast. Every competitor conditions on the same information set at t, but the embargo leaves the ridge with h fewer training observations, an asymmetry that works against it and not in its favour.

Accuracy is measured by skill, the percentage reduction in forecast error relative to the random walk,

$$\text{skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{RW}}}, \tag{3}$$

positive when the model wins. We do not report R^2: the series are so persistent that the random walk itself attains $R^2 \approx 0.99$, which makes the statistic flattering and uninformative.

3.2 Predictability exists, but belongs to no model

Table 2: Skill against the random walk (%) on the 30-day implied volatility, out-of-sample 2016–2023. Best model per horizon in bold.

h (trading days)	1	2	5	7	14	30
ARMA	**1.8**	**−1.3**	**0.7**	**2.3**	6.5	13.1
HAR	1.4	−1.7	−0.3	0.8	5.9	12.8
VAR(2)	0.1	−3.3	−0.7	1.4	6.3	13.4
Ridge	0.2	−1.7	−0.0	1.2	**6.7**	**13.9**

Two facts stand out, and the second matters more. Predictability grows with the horizon, from nothing at a day or a week, where all four models are worse than the random walk at two days, to 13 or 14% at a month. But no model class wins: at thirty days the four specifications span 1.1 percentage points, at short horizons the ranking reverses in favour of the simplest, and the non-linear alternatives of Appendix A do strictly worse. Whatever is forecastable here is a property of the series, not a reward for modelling effort.

One caution changes how the last column should be read. On the test window the standard deviation of IV_{30} is 7.54 while the random walk's error at $h = 30$ is 7.68, so predicting the unconditional level would already do better than predicting no change. Table 3 settles the question with two anchored benchmarks, an expanding-window mean and an AR(1) reverting to that mean at rate $\hat{\rho}_t^h$, both estimated walk-forward.

Table 3: Root mean squared error of the ridge and of three benchmarks, and the implied skill (%). All four forecasters are scored on the index common to them, which is why the skill against the random walk differs from Table 2 by a tenth of a point.

	RMSE of each forecaster (vol points)				Skill of the ridge against		
h	RW	Mean	AR(1)	Ridge	RW	Mean	AR(1)
5	3.701	7.560	3.843	3.706	−0.1	51.0	3.6
14	5.879	7.586	5.910	5.491	6.6	27.6	7.1
30	7.677	7.626	7.082	6.620	13.8	13.2	6.5

The answer is partly yes. The unconditional mean is a poor forecaster at five and fourteen days, so the monthly result is not merely a level effect. But the anchored AR(1) is a much stronger competitor at a month, and against it the ridge retains only 6.5%. **Roughly half of**

the apparent monthly skill is mean reversion that a two-parameter model already captures. We report skill against the random walk throughout for comparability with the literature, but the monthly headline should be read as 6.5%.

3.3 How solid is it?

Three checks discipline these numbers, and each weakens the claim.

The random walk is nested in every competitor, so the Diebold–Mariano statistic is mis-sized and the appropriate test is that of Clark and West (2007). The Newey–West truncation is where the answer lives: at the automatic rule $\lfloor 4(T/100)^{2/9} \rfloor = 7$ the ridge reaches 2.83 at $h = 30$, but seven lags cannot span a thirty-day overlap, and at 29, 45 and 60 lags the statistic falls to 2.05, 1.98 and 1.95, moving the one-sided p-value from 0.002 to about 0.025. We use $\max(2h, 7)$ throughout, and at that truncation the monthly statistics are 1.94 (ARMA), 1.98 (HAR), 1.85 (VAR) and 1.95 (ridge), all within 0.13 of one another. All specifications reject the null, though narrowly, the one-sided p-value being close to 0.025.

That last sentence would once have closed the matter, and it should not. Under a recursive estimation scheme with parameter estimation error, the limiting distribution of a multi-step Clark–West statistic is not that of the one-step case (West, 1996; Clark and McCracken, 2005), so the normal reference is an approximation of unestablished quality at $h = 30$. We therefore replace it by two references that do not depend on it, reported in Table 4.

The first is a block bootstrap. The adjusted loss whose mean defines the statistic has zero expectation under the null, so recentring it and resampling it in moving blocks of $\max(2h, 7)$ generates the statistic under the null without any distributional assumption; bootstrap references for nested predictive comparisons of this kind are studied by Clark and McCracken (2012). We studentize throughout by the Newey–West standard error of the original sample rather than recomputing it inside each replication: a reblocked sample has broken dependence at its joins, the Newey–West estimator is biased down by some eighteen percent there, and the resulting ratio has a degenerate distribution (standard deviation 3.7 but a ninety-fifth percentile of 1.26). With the original standard error the bootstrap distribution has standard deviation 1.00 to 1.05 at every horizon, which independently validates the block length, and what is then tested is the *shape* of the null, which is exactly what is in question. That shape deteriorates with the horizon precisely as the theory warns: the skewness of the bootstrap null rises from 0.03 at five days to 0.9 at thirty, and the five percent critical value rises from 1.71 to 2.02 against the normal's 1.645. The consequence is a reversal. At fourteen days the four statistics remain significant ($p_B = 0.040$ to 0.044). At thirty days none of them is: p_B runs from 0.051 to 0.069 where the normal gave 0.024 to 0.032.

The second reference removes the overlap rather than modelling it. Taking every h-th observation yields h subseries with no overlapping errors at all, on which one-step theory applies unaltered. At thirty days these are thirty subseries of sixty-six observations each; their median t is 1.80 to 1.95, close to the full-sample value, but only sixty to seventy-seven percent of them exceed 1.645 and their range is wide, from 1.13 to 3.68. The point estimate is therefore not an artefact of the overlap; its significance at the monthly horizon is.

The second check is the Model Confidence Set of Hansen, Lunde and Nason (2011), which returns the specifications that cannot be distinguished from the best. The verdict is uniform: at no horizon can any model be distinguished from the random walk, the procedure stopping at its first elimination test with $p = 0.94$, 0.38 and 0.12 at five, fourteen and thirty days. That is a statement about power, not about equality: with a block of sixty on nineteen hundred observations the resampling has some thirty effective units, and a procedure with thirty units eliminates little whatever the truth. The outcome turns on one implementation detail. The block must span the window over which the errors overlap, and the conventional $\lfloor n^{1/3} \rfloor = 12$ observations does not at $h = 30$; with a block of twelve the random walk is excluded at $p = 0.057$, and with $\max(2h, 5)$ nothing is eliminated anywhere. We use the latter throughout, with 2,000

Table 4: Bootstrap reference for the multi-step Clark–West statistic. Column p_N refers the statistic to the standard normal; p_B refers it to the distribution of the same statistic on 2,000 moving-block resamples of the recentred adjusted loss, that is under the null, studentized throughout by the Newey–West standard error of the original sample. The last three columns split the sample into h non-overlapping subseries taken every h days, for which one-step theory applies without correction, and report the median t, its range across subseries and the share exceeding 1.645.

h	model	t	p_N	p_B	q^B_{95}	med. t_{sub}	range	share
5	ARMA	1.81	0.035	0.043	1.71	1.45	0.67–2.05	20%
	HAR	1.44	0.075	0.081	1.68	1.32	0.62–1.56	0%
	VAR	1.97	0.025	0.035	1.69	1.93	1.11–2.55	80%
	Ridge	1.57	0.058	0.070	1.77	1.30	0.65–1.92	20%
14	ARMA	2.11	0.017	0.041	1.95	1.75	0.99–3.24	64%
	HAR	2.01	0.022	0.043	1.92	1.51	0.86–3.58	36%
	VAR	1.97	0.024	0.044	1.90	1.60	0.96–3.58	50%
	Ridge	2.02	0.022	0.040	1.87	1.60	0.93–3.64	50%
30	ARMA	1.94	0.026	0.051	1.94	1.93	1.17–3.34	77%
	HAR	1.98	0.024	0.055	2.02	1.95	1.14–3.68	77%
	VAR	1.85	0.032	0.069	2.01	1.80	1.13–3.50	60%
	Ridge	1.95	0.026	0.051	1.96	1.87	1.14–3.57	63%

replications of a moving block bootstrap in the sense of Künsch (1989), that is with blocks of fixed length and random starting points. The stationary variant of Politis and Romano (1994), which draws block lengths from a geometric distribution so that the resampled series is itself stationary, would be the more careful choice; we did not find the distinction material here, but we have not established that it is immaterial and the label should be read literally.

The third is sampling uncertainty. Forecasting a thirty-day change every day produces heavily overlapping observations, so two thousand daily forecasts amount to roughly sixty-five independent ones, and the bootstrap interval for the 13.9% skill at $h = 30$ is accordingly wide, (3.9, 31.4); at fourteen and five days it contains zero.

The size of the family also has to be declared rather than left implicit. We fix it as the union of the two tables we report, the twenty-four tests of Table 2 and the eighteen of Table 5, for forty-two Clark–West statistics. Thirty-seven reject individually at five percent and seven survive Holm's family-wise correction. Benjamini–Hochberg retains thirty-seven, but that figure should not be quoted on its own: our statistics are strongly dependent, since the horizons are nested and the h-step errors overlap, and Benjamini–Hochberg controls the false discovery rate only under independence or positive regression dependence. Under arbitrary dependence the Benjamini–Yekutieli correction applies, whose penalty here is $\sum_{i\le 42} 1/i = 4.33$; it leaves eleven. The three counts, seven, eleven and thirty-seven, bracket what the family can support, and the composition matters more than any of them: five of the seven Holm survivors and five of the eleven under Benjamini–Yekutieli are the curvature. We do not pretend this family is the whole search. The specifications relegated to Appendix A, the penalty and decay grids, the two curve constructions and the two capture times all belong to the effective space we explored, and a full accounting would be larger and less favourable.

None of these three corrections exploits the dependence between our tests, which is strong because the horizons are nested and the h-step errors overlap; each instead pays a penalty for a multiplicity that is partly illusory. The stepdown procedure of Romano and Wolf (2005) does exploit it. We resample the forty-two adjusted losses *jointly*, in common moving blocks of sixty so that the cross-test dependence is carried into every replication, studentize each by its own $\max(2h, 7)$ Newey–West standard error, and step down through the ordered statistics

against the bootstrap distribution of their maximum. Because that maximum is far smaller than the maximum of forty-two independent statistics would be, the procedure is markedly less conservative: it retains **seventeen** of the forty-two, against seven for Holm and eleven for Benjamini–Yekutieli, while controlling the same family-wise error rate. It also uses a bootstrap null rather than a normal one, so it corrects the multiplicity and the approximation of the previous paragraph at once.

The composition of those seventeen is what matters, and it does not rescue the headline. All six curvature tests survive. Not one of the four monthly forecasts of the thirty-day volatility does: their adjusted p-values lie between 0.061 and 0.063, and the bootstrap p-values of Table 4 lie between 0.051 and 0.069. Two procedures built on different principles, one correcting the reference distribution and the other the multiplicity, place the same four statistics in the same narrow band just above five percent.

We do not pretend even this family is the whole search, and the same logic applies to the economic metric, where the space we explored is a grid of thresholds and risk targets rather than a list of models. Section 5 therefore applies Hansen's (2005) test of superior predictive ability to that grid directly.

The defensible reading is therefore narrow. Monthly-horizon predictability is a large point estimate with a wide interval, obtained on few independent observations, that cannot be attributed to any model, halves against a properly anchored benchmark, and loses significance under both a bootstrap reference and a dependence-aware multiplicity correction. We treat it as suggestive at best and rest the paper on findings that do not depend on it.

3.4 Where the little edge lives

Decomposing by factor explains the uniformity of Table 2. On the level the two models are close, the ARMA leading at five days and the ridge catching it by a month, which is what one expects of a nearly purely autoregressive series. On the slope they are close throughout. The ridge's advantage is concentrated, and large, on the curvature.

Table 5: Skill against the random walk (%) by factor. The advantage of conditioning on the shape of the curve is confined to curvature.

	β_0 (level)		β_1 (slope)		β_2 (curvature)	
h	ARMA	Ridge	ARMA	Ridge	ARMA	Ridge
5	**4.7**	2.0	**1.5**	0.4	3.7	**5.5**
14	**5.2**	4.9	8.9	**10.1**	6.1	**20.8**
30	5.5	**5.9**	18.5	**19.4**	6.4	**26.4**

That edge does not matter, and the reason is arithmetic. A factor's marginal contribution to the standard deviation of the fitted volatility at a given maturity is its loading there times its own standard deviation, ignoring the covariances between factors, so what follows is an accounting of magnitudes rather than a variance decomposition. At thirty days the slope contributes 5.08 vol points and the level 3.03, but the curvature only 1.04, and never more than 1.88 anywhere on the curve, because its loading peaks at 0.30 at about 122 days. The advantage of conditioning on the shape sits on the factor that matters least, which is why a 26% edge on β_2 buys at most a one-point difference in Table 2.

Put beside Section 3.3, this is the sharpest thing the statistical half of the paper has to say. Of the seven effects that survive Holm's correction out of forty-two, five are on the curvature: the ARMA at five, fourteen and thirty days and the ridge at five and fourteen, with corrected p-values from below 10^{-4} to 0.038. And the curvature is the one factor that cannot move the traded quantity. **The predictability that survives our own multiplicity correction is concentrated on the component that is economically inert.**

One diagnostic belongs here, because concentration in a few days is the usual failure mode of a result of this kind. The fifty most influential days out of roughly nineteen hundred account for 78% of the curvature's measured gain at fourteen days, which sounds damning until the same statistic is computed for the rest of Table 5: it exceeds 100% in every other cell, meaning the remaining days are collectively negative there. Squared-error comparisons on heavy-tailed series concentrate by construction, so three checks that do not depend on squaring are more informative. Measured in mean absolute error the curvature skill is 16.9% and 19.1% at fourteen and thirty days, against 20.8% and 26.4% in root mean squared error; the ridge has the smaller absolute error on 56.3% and 54.6% of days, the highest count anywhere in the table; and deleting the fifty most influential days leaves +6.5% and +9.2%, where every other cell turns negative. The curvature result is therefore the most robust in Table 5, not the least. Its five-day figure does not survive the same checks and we do not lean on it.

Diffuseness alone, however, does not separate genuine predictability from a mechanical artefact, and one artefact is specific to this setting. Write the estimated factor as $\beta_{2,t} = \beta^*_{2,t} + e_t$, where e_t is the error of the daily cross-sectional fit. The random walk carries e_t into its forecast and pays for it twice, once at t and once at $t+h$, while a model conditioning on smoother inputs pays only once. That produces skill against the random walk even with no ability to predict β^*_2, and it grows with the noise-to-signal ratio of the factor. We can measure it. Fitting Equation (2) separately on the odd- and even-ranked expiries of each day yields two nearly independent readings of the same factor, whose correlation is the reliability ratio: 0.970 for the level, 0.983 for the slope and 0.828 for the curvature, which confirms that the curvature is by far the noisiest of the three and that the permissive spread filter of Section 2 shows up here. From that reliability we recover the measurement variance and, with the variance of the true h-day change, the skill the mechanism alone would generate. It is 4.7% at fourteen days and 4.1% at thirty, against 20.8% and 26.4% observed. Errors in variables therefore account for something like a fifth of the curvature result and do not explain it away. Two biases run in opposite directions and we cannot sign the net: the calculation credits the model with carrying no measurement error at all, which inflates the mechanical part, while the odd-even split cancels any error common to a whole day's chain, which deflates the measurement variance and hence the mechanical part. The figures are an order of magnitude rather than a bound. We report them because the mechanism is real and, to our knowledge, not usually checked.

A duller reading of the whole is nonetheless available, since the curvature is also the least persistent of the three factors and therefore the one where the random walk is easiest to beat for reasons that have nothing to do with the economics. What these facts jointly establish is weaker than the display above but still worth stating: nothing that survives our correction is large enough to move a traded price.

4 How would we trade it?

4.1 Is the forecast distinct from the price?

Beating a random walk is a statistical achievement, but beating the market's own forecast is an economic one, and the curve already contains a forecast. If today's volatility is 15% at thirty days and 17% at sixty, the market is implicitly quoting a volatility for the window between them, given by the variance identity

$$\sigma^2_{\mathrm{fwd}}(h) = \frac{\sigma^2(h+\tau)\,(h+\tau) - \sigma^2(h)\,h}{\tau}. \tag{4}$$

The gap between that forward and today's level is the carry, what an investor earns or pays if the curve stays put, and a forecast worth trading must beat it. The identity is exact for variance-swap rates and only approximate for the at-the-money volatilities we work with, so the carry should be read as an approximation to the market's forward. One practical concern does

not materialize: the implied forward variance is positive on all but one of the 3,354 days at every horizon we use, including the March 2020 inversion.

Regressing the realized change on the model forecast and on the carry cannot answer the question. The two predictors correlate around 0.90, so the coefficients are unstable, and both are computed from the same fitted curve, so their measurement errors are correlated by construction. We replace that regression by two tests that are not compromised. The first makes the forward compete directly: Table 6 enters it as a fourth forecaster and re-runs the model confidence set.

Table 6: Model confidence set at 90% with the curve's own forward volatility entered as a competing forecaster. Root mean squared errors in volatility points, lowest at each horizon in bold. Block length $\max(2h, 5)$. Nothing is eliminated at any horizon.

h	RW	Ridge	HAR	Forward
1	2.054	2.050	**2.024**	2.034
2	2.423	2.465	2.464	**2.395**
5	3.704	3.706	3.714	**3.602**
7	4.320	4.269	4.280	**4.163**
14	5.884	**5.491**	5.538	5.569
30	7.685	**6.620**	6.703	6.936

From two to seven days the forward has the lowest error of the four; at fourteen and thirty the ridge does. At no horizon can any of them be excluded. The defensible conclusion is not that the carry spans our forecast, which the collinearity makes untestable, but the weaker one: **the market's own quote is at least as good a forecast as anything we build out to a week**.

The second test runs the regression in traded space, on quantities the market quotes rather than quantities we fit. Table 7 regresses the realized change in the constant-maturity futures price on the model's predicted change in that price, obtained by the projection of Section 4.2, and on the futures basis.

Table 7: Spanning regression in traded space. Heteroskedasticity- and autocorrelation-consistent (HAC) t-statistics at $\max(2h, 7)$ Newey–West lags, referred to the standard normal; bold marks $|t| > 1.96$. The last column is the correlation between the two regressors, which is around 0.90 in spot space.

h	b_{model}	t_{model}	b_{basis}	t_{basis}	R^2	n	ρ
1	0.057	1.19	0.018	0.40	0.006	1891	−0.45
2	0.109	1.29	−0.012	−0.13	0.019	1890	−0.38
5	0.229	1.86	0.015	0.09	0.034	1887	−0.21
7	0.271	**2.32**	0.053	0.26	0.041	1885	−0.13
14	0.412	**3.49**	0.108	0.41	0.085	1878	0.04
30	0.556	**4.48**	0.300	1.57	0.178	1862	0.21

The collinearity falls away once the objects are expressed in the units of the quote. Up to five days neither regressor is significant; from seven days the model forecast is, rising to $t = 4.48$ at thirty, while the basis never reaches significance. Read naively this is identifiable predictive content in the forecast for the traded price at a week and beyond, and not the basis in disguise. It was the only positive claim in an otherwise negative paper. It does not survive scrutiny, and Table 8 is the demonstration.

The design manufactures the result. The regressand is $F_{t+h} - F_t$ and both regressors contain F_t, the first with a negative sign. Nothing about that is innocuous, because F_t is very persistent: its first-order autocorrelation is 0.982, implying a half-life of thirty-nine days, and

its correlation with its own value twenty days later is still 0.738. A persistent series produces a negative relation between its level and its subsequent change over any horizon comparable to that half-life. Since $-F_t$ sits inside the regressor, that relation appears as a *positive* coefficient on $\hat{F}_t - F_t$ whether or not $\hat{F}_t$ contains any information at all. Three tests separate the two possibilities, and all three give the same answer.

The first frees the restriction. Writing the regression with $\hat{F}_t$, F_t and IV_t as separate regressors nests the reported specification exactly, and imposes one restriction fewer. The t-statistic on the forecast is then **0.10** at thirty days and negative at every shorter horizon, and the R^2 moves from 0.178 to 0.184: the entire explanatory power, and slightly more, comes from the two levels. Dropping the forecast altogether leaves an R^2 of 0.184, identical to three decimals.

The second substitutes an object that cannot forecast anything. Replacing $\hat{F}_t$ by the expanding mean of F up to t yields $t = 4.00$ and $R^2 = 0.179$ at thirty days, against the model's 4.48 and 0.178. A quantity with no information about the future reproduces the result to within rounding.

The third asks directly whether the design can manufacture $t = 4.48$ under a null in which the forecast is uninformative but everything else is intact. We shift the forecast series by a random circular offset, which leaves its marginal distribution, its autocorrelation and the whole $-F_t$ structure untouched while destroying its alignment with the data, and refit. Over 2,000 shifts the mean t-statistic is 3.41 and the ninety-fifth percentile is 4.43. The observed 4.48 sits just past that boundary, $p = 0.041$: three quarters of the statistic is mechanical, and what remains is at the edge of significance before any correction for the fact that this is one test among forty-two.

We therefore withdraw the claim. The regression in traded space does not establish predictive content; it establishes that the constant-maturity futures price mean-reverts, which is not news and is not ours. Consistently with this, in the return space of the next paragraph, where the regressand no longer contains F_t, the unrestricted coefficient on the forecast is negative ($t = -1.44$ at thirty days).

Table 8: Is the spanning result in traded space mechanical? Columns 2–4 reproduce the restricted specification, which imposes a single coefficient on $\hat{F}_t - F_t$. Column 5 frees that restriction and reports the t-statistic on $\hat{F}_t$ when F_t and IV_t enter separately; column 7 drops the forecast entirely and keeps only those two levels. Column 8 replaces the forecast by the running mean of F, an object containing no information about the future. The last two columns give the mean and the exceedance probability of the restricted t-statistic over 2,000 random circular shifts of the forecast series, which destroy its alignment with the data while preserving its marginal distribution, its autocorrelation and the appearance of F_t on both sides. HAC standard errors at $\max(2h, 7)$ lags throughout; bold marks $|t| > 1.96$.

	restricted			unrestricted		levels	placebo	shift test	
h	b	t	R^2	$t_{\hat{F}}$	R^2	R^2	t	$\bar{t}$	p
1	0.057	1.19	0.006	−0.23	0.009	0.009	1.87	1.92	0.970
2	0.109	1.29	0.019	−0.25	0.026	0.025	**2.08**	1.97	0.943
5	0.229	1.86	0.034	−0.15	0.043	0.042	**2.65**	2.36	0.770
7	0.271	**2.32**	0.041	−0.21	0.052	0.051	**3.05**	2.54	0.593
14	0.412	**3.49**	0.085	−0.18	0.095	0.095	**3.39**	2.83	0.058
30	0.556	**4.48**	0.178	0.10	0.184	0.184	**4.00**	3.41	0.041

Even had the coefficient been genuine, there would have been a second trap in reading it as an economic result, and it is worth keeping because it is a distinct point. An R^2 of 0.178 on a thirty-day change, if it were a return, would imply a Sharpe near 1.2, nowhere near what Section 5 finds. The constant-maturity price and the return of a constant-maturity position are not the same object: holding the position earns the weighted returns of two contracts that each slide down the curve as they age, while the interpolated price always looks at the same thirty-day point and never records that slide. Over our window the interpolated price drifts by −3.9% a year while the position returns −47.4%, and the gap is the roll. Re-running the regression on

the cumulative return over h days, the quantity actually monetized, leaves only the thirty-day coefficient marginally significant ($t = 2.00$) and collapses the R^2 to 0.020, which implies a Sharpe near 0.4 before costs and is consistent with the backtest. Statistical identification and economic value are different questions, and the distance between them here is measured by the roll.

4.2 The instrument and the rule

The only liquid, directly tradable claim on the thirty-day point of the curve is the VIX future. We build a constant-maturity position from CFE settlement prices, time-weighting the two contracts bracketing thirty days and rolling daily. It behaves as one expects of short-dated volatility exposure: 76.6% annualized volatility over 2013–2023, a mean return of -38.8% a year from the roll drag of a curve usually in contango, and a worst day of -26.1% on 6 February 2018. On the test window the drag is steeper, at -47.4%.

The gap between what we forecast and what we trade must be quantified rather than asserted. Regressing the futures price F_t on our thirty-day at-the-money volatility gives $F_t = 7.26 + 0.809\,\mathrm{IV}_{30,t}$ with $R^2 = 0.884$ and $n = 1{,}897$: the two track each other closely in variation but are separated by a persistent level gap of $+4.27$ points, positive on 97% of days. It would be a mistake to read that gap as the volatility risk premium. The VIX is a variance-swap rate computed across the entire strike range, so it embeds the deep out-of-the-money puts that carry the index skew, whereas our target is a single at-the-money volatility; on the S&P 500 that definitional wedge is worth two to three volatility points on its own, against a contango closer to one and a half.

The consequence is not cosmetic. A rule comparing a forecast in at-the-money units against a price in variance-swap units would inherit the wedge as a permanent short bias: against a median forecast edge of 0.9 to 1.7 points, the signal would be negative on almost every day and the rule would test nothing. We therefore map the forecast into the space of the traded price before signing it, using the same regression re-estimated walk-forward on an expanding window,

$$\widehat{F}_{t+h} = \hat{a}_t + \hat{b}_t\,\widehat{\mathrm{IV}}_{t+h}, \qquad (\hat{a}_t, \hat{b}_t) \text{ fitted on information up to } t, \tag{5}$$

with $\hat{a}_t$ ranging over [6.37, 8.11] and $\hat{b}_t$ over [0.627, 0.823]. The wedge is absorbed into the intercept, the residual basis is centred by construction, and the rule becomes genuinely two-sided.

One property of this mapping should be stated rather than left for a reader to find. It is estimated on levels, while the quantity that separates the model from its placebo is $\hat{b}_t(\widehat{\mathrm{IV}}_{t+h} - \mathrm{IV}_t)$, a scaled *change*. The two coefficients differ: regressing changes in F on changes in IV_{30} over the same window gives 0.49 at one day rising to 0.65 at thirty, against 0.809 in levels, so Equation (5) magnifies the model's departure from the carry trade by about a quarter. Re-running the whole exercise with a walk-forward change-regression in place of the level one leaves the paired difference at -0.14 and -0.16 at fourteen and thirty days, against -0.14 and -0.17, and moves it to -0.23 ($t = -1.33$) at five days. The conclusion is unaffected and slightly strengthened, so we keep the level mapping for its transparency and record the alternative here.

The rule itself is deliberately simple. The raw signal is ternary, with a dead zone that prevents trading on noise,

$$w_t^{\mathrm{raw}} = \mathrm{sign}\big(\widehat{F}_{t+h} - F_t\big) \cdot \mathbf{1}\big\{|\widehat{F}_{t+h} - F_t| > \delta\big\}, \qquad \delta = 0.3 \text{ vol points}, \tag{6}$$

and it is then scaled so that the position carries a constant amount of risk,

$$w_t = w_t^{\mathrm{raw}} \cdot \min\left(\frac{\sigma^*}{\hat{\sigma}_{t-1}^{(20)}},\ \ell\right), \qquad \sigma^* = 15\%, \quad \ell = 2, \tag{7}$$

where $\hat{\sigma}_{t-1}^{(20)}$ is trailing twenty-day realized volatility and the leverage cap is never reached. The strategy return is $w_{t-1} r_t - c\,|w_t - w_{t-1}|$ and positions are never clipped, so tail losses appear in

full. We also report a *spot-referenced* variant that signs the forecast against IV_t instead of F_t, which is what a purely statistical reading of Section 3 would suggest and which is mis-specified in an instructive way.

The benchmark is the same rule with the model removed. Our random-walk placebo replaces $\widehat{\mathrm{IV}}_{t+h}$ by IV_t and changes nothing else, keeping the same reference price, threshold, risk targeting, costs, days and projection, so that its signal is the negative of the residual basis. It is the nested special case of the strategy in exactly the sense that the random walk is the nested special case of the forecasting models, and the difference between the two isolates the contribution of the model. We also report passive short volatility, the unconditional position. On costs, the minimum tick on VIX futures is 0.05 index points, about 33 basis points of notional on a VIX near 15; we take 33 bp as the central case and 17 bp as an optimistic bound. Execution costs of the daily roll are omitted from strategy and benchmarks alike, so they affect no comparison.

5 Do we make money?

Table 9: Trading the forecast on 30-day constant-maturity VIX futures, out-of-sample 2016–2023, net of $c = 33$ bp and gross of roll costs. The forecast is mapped into futures units by Equation (5) before being signed; the placebo is the identical rule with a random walk substituted for the model. Brackets are HAC standard errors of the annualized Sharpe ratio (Lo, 2002); "Long" is the share of days with a positive position. The rows are scored on 1,886, 1,877 and 1,863 daily observations, and all t-statistics are referred to the standard normal rather than to a t distribution, the influence-function estimator being asymptotic. Two benchmarks for comparison. Passive short volatility earns 0.30 [0.40] over the whole window and 0.27, 0.28 and 0.26 on the days of each row, with a CAGR of 4.0% and a maximum drawdown of −51.9%. The spot-referenced rule, which signs the forecast against IV_t instead of F_t, earns −0.43, −0.37 and −0.15.

h	Model	Placebo	Model − Placebo	t	CAGR (%)	Max DD (%)	Long
5	0.35 [0.36]	0.44 [0.33]	−0.09 [0.19]	−0.45	5.6	−40.7	20%
14	0.31 [0.36]	0.45 [0.32]	−0.14 [0.26]	−0.53	4.8	−43.8	26%
30	0.29 [0.36]	0.46 [0.32]	−0.17 [0.29]	−0.61	4.3	−43.8	33%

Once the forecast is projected into futures units the position is long on 20 to 33% of days, so the rule is testing the forecast rather than inheriting a definitional short bias. It earns a Sharpe ratio of 0.29 to 0.35 before roll costs, with a compound return of 4 to 6% and a drawdown near −44%. Two qualifications matter more than that point estimate.

The level is not distinguishable from zero. A Sharpe of 0.29 carries a standard error of 0.36, so the 95% interval runs from −0.42 to +1.00. We do not claim that the strategy makes money, only that its point estimate is positive and that we cannot tell it from zero on eight years of data.

The model then contributes nothing we can measure, and if anything it subtracts. Replacing the forecast by a random walk, keeping the identical rule, gives 0.44, 0.45 and 0.46, above the model at all three horizons; the paired differences are −0.09, −0.14 and −0.17 with t-statistics of −0.45, −0.53 and −0.61. The pairing is what gives this test what power it has, since the two strategies hold overlapping positions and their difference is far less noisy than either series: the standard error of the difference is 0.19 to 0.29 against 0.36 for the levels, and a stationary block bootstrap gives 0.24 to 0.31. The bound follows directly. The ninety-five percent upper limit on the forecast's contribution is +0.28, +0.37 and +0.40 of Sharpe under the HAC standard errors, and +0.38 to +0.44 under the bootstrap. At every horizon that limit is of the same order as the strategy's own Sharpe ratio, so the sample excludes a large contribution and not a moderate one. That is considerably weaker than "the forecast is worthless", and it is what the data support.

The bound rests on that standard error, so the interval had better be the width it claims to be on returns as badly behaved as these, whose skewness is above ten and kurtosis near three

hundred. We resample the two arms' realized daily returns jointly in blocks, having demeaned both so that the true difference in Sharpe ratios is zero by construction, and count how often the HAC interval covers it. Coverage is 96.6%, 95.6% and 96.7% for blocks of five, twenty-one and sixty-three days against a nominal 95%, and the average standard error of 0.30 matches the empirical dispersion of the resampled differences to within four percent. The interval is correctly sized, marginally conservative if anything, so the bound is a measurement and not an assertion.

A null paired difference would be uninteresting if the two rules simply held the same position, so their disagreement must be reported rather than assumed. The two signals differ by a median of 0.52, 0.94 and 1.47 volatility points, they produce different positions on 30%, 40% and 44% of days, and they take outright opposite positions on 5%, 13% and 22%. At a month the two strategies therefore disagree on nearly half the sample, so the vanishing difference in their Sharpe ratios is a result rather than an artifact; at five days the overlap is high, the position correlation reaching 0.67 against 0.34 at a month, and the bound there should be read with more caution.

The days of disagreement also say where the forecast fails. On those where the two rules take opposite positions, which is where the forecast overrides the basis, the model contributes -9.1, -21.7 and -28.2 percentage points of annualized return, partly offset by the days on which the two agree and differ only in size. The forecast is therefore not uninformative in the neutral sense: the positions it takes against the basis are, in this sample, the wrong ones, and the spot-referenced rule points the same way. None of these figures is individually significant, so we stop short of calling the signal wrong-signed.

Does the placebo itself beat doing nothing? Its signal is the negative of the residual basis, so it is a basis trade of the kind Simon and Campasano (2014) document as profitable, and it earns 0.44 to 0.46 against 0.26 to 0.28 for the unconditional short position on the same days. The paired test cannot resolve that gap: the standard error is 0.59, larger than either individual one. The reason qualifies our own method. Pairing buys precision only when the two arms hold correlated positions, as the model and the placebo do; the placebo and the passive position are *negatively* correlated at -0.27, because the placebo goes long whenever the residual basis is negative while the passive is always short. Eight years cannot tell a basis trade from an unconditional short, and our method offers no help here.

Four design choices could conceivably hide the information that Section 4 identifies, and none of them does. Table 10 reports the paired difference under each; the levels move a great deal, halving at the longer horizons once execution is delayed, but the difference stays between -0.24 and $+0.07$ with t-statistics below 0.9 in absolute value. The trading threshold is not hiding anything either: varying δ from zero to 1.2 volatility points, four times our setting, moves the thirty-day difference over -0.14, -0.17, -0.32 and -0.20, negative throughout and never reaching $|t| = 1.1$. Splitting the test window at $h = 30$ gives -0.27 over 2016–2019 and -0.04 over 2020–2023, neither significant. Signing the raw at-the-money forecast against F_t without the projection gives differences of $+0.04$, -0.02 and -0.02 with standard errors of 0.13 to 0.17, but both legs are then long on at most one percent of days and reduce to the same unconditional short, so the apparent precision belongs to a degenerate design and we decline to borrow its confidence interval.

Roll costs reduce the profit without removing it, and the arithmetic has to be done at the position actually held. Maintaining a constant thirty-day maturity shifts weight between the two bracketing contracts every day, 25.1 units of notional a year. But the rule never holds a unit of exposure: volatility targeting at 15% against an instrument whose volatility is 76.6% leaves a mean absolute position of 0.254 at thirty days. Scaled to the positions held, the roll is 6.4 units a year, costing 1.1% at 17 bp and 2.1% at 33 bp against a gross CAGR of 4.3%, so the strategy compounds at 3.2% or 2.2% net. Because the charge is close to a constant drag it lowers the numerator of the Sharpe ratio without touching the denominator, taking it from 0.29 to 0.24 at 17 bp and 0.20 at 33 bp, so the net figure stays positive and stays indistinguishable from zero. The paired difference is untouched at -0.18. The charge is structural rather than an artifact of

Table 10: Paired difference in Sharpe ratio, model minus placebo, under four specifications of the rule. *Proportional* replaces the ternary signal by a position proportional to the standardized edge, clipped at two standard deviations, with no dead zone; *full-horizon* holds each signal for h days in overlapping tranches; *lagged* delays execution by a full trading day. Every entry has $|t| < 0.9$.

h	5	14	30
Baseline	−0.09	−0.14	−0.17
Proportional	+0.07	+0.03	−0.03
Full-horizon	−0.08	−0.06	−0.07
Lagged one day	−0.08	−0.21	−0.24

the constant-maturity construction, since holding the nearest contract beyond ten days to expiry turns over 24.0 units a year against 25.1; and since it falls in the same proportion on all three portfolios, no comparison in Table 9 is affected.

Referenced to spot volatility instead, the same forecast loses, returning −0.43, −0.37 and −0.15. That is what a purely statistical reading of Section 3 would lead one to trade. The contrast is large, but it would be wrong to read it as a clean measure of what the referencing choice costs, and a decomposition says why. Writing the gross annual return as $\mathbb{E}[w]\,\mathbb{E}[r] + \mathrm{Cov}(w, r)$ separates what a rule earns from its average exposure to the instrument from what it earns by varying that exposure. It decomposes a mean and not a ratio, so it explains a difference in Sharpe ratios only when the two arms carry comparable risk; here they do, at 22.7% and 22.3% of annualized volatility, and the gap therefore sits in the numerator. At thirty days the projected rule earns +0.1 and +9.7 percentage points from the two terms, its mean position being −0.002: it is flat on average and is therefore not a disguised short-volatility position. The spot-referenced rule earns −4.3 and +5.8, its mean position being +0.095. Its loss is a net long exposure to an instrument that drifts down at 47% a year, not an absence of timing. Referencing a forecast against spot rather than against the traded price is a real error, and one the forecasting literature does commit, but what it costs here is mostly a change in average exposure, so the contrast restates the premium rather than adding to it.

The same decomposition sharpens the paired comparison, and slightly qualifies it. The placebo's advantage over the model at thirty days, +4.7 points of gross annual return, is +3.7 from average exposure and only +1.0 from timing: in the aggregate the model does not time worse, it holds a less short book. That is consistent with the loss recorded earlier on the days of opposite sign, which is concentrated on the fifth of the sample where the forecast overrides the basis and is largely offset elsewhere, leaving the covariance term almost unchanged while the mean position moves. Since the placebo's signal is the negative of the residual basis, holding a shorter book on average is precisely harvesting the premium, so what the forecast does is dilute that harvest rather than mistime it. This is the paper's conclusion restated at the level of the position rather than of the Sharpe ratio, and it is the more informative statement of the two.

One refinement guards against reading the timing term as a property of the signal when it might belong to the risk targeting, since deleveraging after a volatility spike is a known source of return in its own right. The same +9.8 points can be split a second way, into what the sign of the ternary signal earns at the average level of scaling and what the covariation between signal and scaling adds. The first term is +10.3 and the second −0.5: volatility targeting slightly reduces what the rule earns rather than generating it. The timing therefore sits in the sign of the signal.

A word finally on the metric. The instrument has an annualized volatility of 76.6% and returns whose skewness is +10.3 and kurtosis 281; the Sharpe ratio is a poor summary of such a distribution, and the influence-function standard error of Lo (2002) assumes finite fourth moments, which these series test severely. We therefore checked that two events do not drive the comparison. Excluding February 2018 the paired difference at thirty days is −0.23, excluding March 2020 it is −0.10, and excluding both it is −0.13, against −0.17 on the full window and

with t-statistics between -0.34 and -0.62 throughout. The sign and the insignificance survive; the magnitude moves by about a third, which is one more reason to read the bound as an order of magnitude.

All of this is consistent with Simon and Campasano (2014) and Johnson (2017). The carry accounts for the predictable component of the change at short horizons, and what the residual basis pays is a premium rather than a forecast error; their profitable basis trade and our unprofitable carry-referenced signal are different objects, a spot-curve forward on one side and a futures basis on the other.

Figure 2 shows the three portfolios cumulated. It is a useful corrective to the tables, because equal or near-equal Sharpe ratios do not imply similar paths: the model and the placebo part company sharply in early 2018 and never fully reconverge, and it is the day-by-day pairing rather than the endpoints that the test compares.

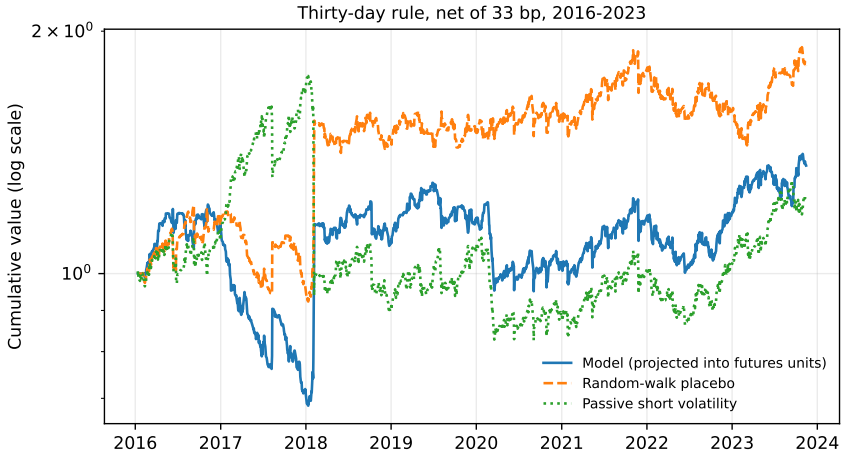


Figure 2: The strategy at $h = 30$ (solid), its random-walk placebo (dashed) and passive short volatility (dotted), log scale, net of 33 bp and gross of roll costs. The model ends at 1.36 times capital against 1.81 for the placebo and 1.25 for the passive position, and falls to 0.69 in early 2018 against 0.92 for the placebo. The paired test compares the two day by day rather than at the endpoint, and cannot separate them.

Does any configuration of the rule beat the placebo? Every number above is reported for one threshold and one risk target. That is a choice, and we made it after looking at the data, so the paired difference at those settings is not the only comparison we effectively ran. The honest object is the whole grid: three horizons, five thresholds from 0.2 to 1.0 volatility points, three annualized risk targets from ten to twenty percent, forty-five configurations in all, each with its own paired placebo run on the same days with the same settings.

Hansen's (2005) test of superior predictive ability is designed for exactly this: it tests the null that no configuration outperforms the benchmark, taking the maximum over the grid as its statistic and bootstrapping that maximum so that the search is priced in. Its recentring removes configurations too poor to be plausibly at the frontier, which makes it more powerful than White's (2000) reality check on the same grid. Applied to the annualized paired excess return over the placebo, with 2,000 moving-block resamples, the statistic is 0.13 and $p_{\mathbf{SPA}} = 0.82$; the reality check gives 0.83. Three of the forty-five configurations have a positive point estimate, the best of them by 0.6 percentage points a year with $t = 0.13$, and the average across the grid is -4.8 points. The conclusion of Section 5 is therefore not an artefact of the settings we chose to report: no configuration of this rule beats its own placebo once the search is accounted for.

6 Discussion and conclusion

The three questions have answers, though two of them are bounds rather than verdicts.

Can we forecast? Not in a way we can establish. The monthly gain is 13 to 14% against a random walk, 13.2% on a raw rather than a fitted target and 6.5% against an AR(1) anchored on the running mean, and no model survives elimination from the confidence set at any horizon once the bootstrap block spans the overlap of the errors. It rests on about sixty-five independent observations and is identical across four very different models, so it belongs to the series and not to any specification. Two further checks remove what was left of its significance. Referred to a bootstrap null instead of a normal one, the monthly Clark–West statistics move from $p = 0.024$–0.032 to $p = 0.051$–0.069, the normal reference having been anti-conservative because the null is right-skewed at long horizons. And a Romano–Wolf stepdown, which exploits the dependence across the forty-two tests and is therefore markedly less conservative than Holm, retains seventeen of them without retaining a single monthly forecast of the thirty-day point. What it does retain is all six curvature tests, the factor too small to move a traded price.

How would we trade it? With a simple risk-targeted rule on VIX futures, signed against the traded price rather than against spot volatility, but only after noticing that up to a week the curve's own forward forecasts as well as the model, so there is little left to trade. Beyond a week, the regression that appeared to identify predictive content for the futures price does not: the price is persistent, it appears on both sides of the equation, an information-free forecast reproduces the result, and freeing the coefficient restriction leaves the forecast with $t = 0.10$.

Do we make money? Not measurably. At thirty days the point estimate is 0.29 to 0.35 of Sharpe, indistinguishable from zero, and the same rule with a random walk in its place does slightly better, mostly by holding a shorter book rather than by timing better. The paired bound runs from $+0.28$ to $+0.40$, so we exclude a large contribution and not a moderate one. Nor is this the artefact of a threshold and a risk target chosen after the fact: across the forty-five configurations of the grid we explored, a test of superior predictive ability against the paired placebo returns $p = 0.82$.

We state the result in the narrowest form the evidence supports, which is not a statement about market efficiency. A single ternary rule on a single instrument cannot test efficiency, and we do not claim to. What we establish is that relative to a nested benchmark using no forecast, this forecast delivers no measurable incremental economic value, and we bound how large that value could still be.

The result refines Guo, Han and Zhao (2014) rather than contradicting them. Their finding of factor predictability survives on an independent fourteen-year sample. What does not survive is the implicit corollary that predictability is exploitable. It is the term-structure carry, priced into the curve and available to anyone who reads it, and the premium attached to it is earned by holding volatility risk rather than by forecasting it, in the sense documented by Bollerslev, Tauchen and Zhou (2009) and Carr and Wu (2009). In that we recover, on a modern sample and with a factor model rather than a reduced form, the conclusion Gonçalves and Guidolin (2006) and Konstantinidi, Skiadopoulos and Tzagkaraki (2008) reached on earlier data.

Several limitations bound the claim. The rule is deliberately crude, a ternary signal on a single maturity, and a richer mapping might extract value that this one cannot. Three things bound how far that objection carries, and it is worth assembling them here because they are scattered through the paper. First, the comparison is *paired*: a rule too blunt to extract value is equally blunt in both arms, so crudeness moves the level of the Sharpe ratio and not the difference, which is the quantity we measure and bound. Second, the parameters of the rule are not load-bearing. The threshold has been varied from zero to 1.2 volatility points, four times our setting, and the risk target and horizon with it, for forty-five configurations over which a test of superior predictive ability returns $p = 0.82$. Third, the mapping has been varied structurally and not only parametrically: replacing the ternary signal by a position proportional to the standardized edge, with no dead zone at all, gives $+0.07$, $+0.03$ and -0.03 (Table 10), and replacing the linear

forecaster by gradient boosting or a random forest makes the forecast worse rather than better (Appendix A). What remains untested is a mapping that allocates across the whole curve rather than one maturity, and we do not claim to have excluded it. The cost model is stylized, though a finer treatment would work against the strategy and not against the placebo, which trades slightly less. We also tested a sixty-day against thirty-day calendar spread, an instrument built to target the slope rather than the level: both arms lose heavily there and the paired difference is again insignificant, but conditioning gives up so much of the carry on that instrument that the comparison carries little information, so we do not rest anything on it. The study covers one market: extending it to the VSTOXX or the VXN would show whether spanning by the carry is a property of the S&P 500 or of volatility term structures generally. And the diagnostic of Table 8 is a warning about a design, not a proof that no design can find content in the traded object. Any regression of a change on a regressor containing the current level inherits the same artefact whenever that level is persistent, and ours did; a specification that models the VIX futures curve directly, rather than forecasting spot volatility and projecting it into futures units, avoids the two changes of units this paper had to make and separates "no signal" from "signal destroyed in translation". Appendix C runs it, on the futures record alone and on a window five years longer, and the answer is the first: the monthly skill falls from 13–14% to 5.3–5.5%, the out-of-sample R^2 for the position's return is negative against its own historical mean, and the rule does not separate from the carry. The absence is a property of the problem rather than of the translation.

The paired placebo made these statements possible, and its limits deserve to be as clear as its benefits. Every unconditional Sharpe ratio here carries a standard error near 0.35, so the sample on its own distinguishes almost nothing; because the model and the placebo hold positions correlated at 0.34 to 0.67 (with strategy returns correlated at approximately 0.68 to 0.86), their paired difference has a standard error of 0.19 to 0.29, and Section 5 shows that interval to be correctly sized. Applied to the placebo against the passive position, whose positions are negatively correlated, the same test has a standard error of 0.59 and is useless. The rule is not ours and is worth stating in closed form: under the asymptotic variance of Jobson and Korkie (1981) as corrected by Memmel (2003), the paired standard error scales with $\sqrt{2-2\rho}$ for small Sharpe ratios, so relative to a single one it is $\sqrt{2} \approx 1.41$ times larger at zero correlation, equal at $\rho = 0.5$ and $\sqrt{0.1} \approx 0.32$ times as large at 0.95. Pairing is therefore worth having above a correlation of about one half and harmful below that threshold, becoming increasingly so as it turns negative, which is exactly why the nested null is the right benchmark and an arbitrary one is not. Run the null model through the whole pipeline, and report the correlation between the arms so the reader can judge what the pairing is worth.

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A Additional checks

Order selection. BIC on the training window selects $(2, 0, 1)$ for IV_{30} and the slope, $(1, 0, 2)$ for the level and $(1, 0, 1)$ for the curvature, and those orders are held fixed out of sample. Fixing all four at the parsimonious $(1, 0, 1)$ instead raises the ARMA skill on IV_{30} at one day, from 1.8% to 3.8%, and lowers it at thirty, from 13.1% to 12.3%, without changing any conclusion: the model confidence set still eliminates nothing.

Non-linear specifications. Replacing the ridge by two tree ensembles on the same features, targets and walk-forward protocol helps neither. Gradient boosting is worse at every horizon, reaching 5.9% at thirty days and -8.1% at one, and the random forest reaches only 3.5% at thirty days against 12.8 to 13.9% for the linear models. The mapping from curve shape to future volatility is essentially linear, and added flexibility captures noise.

A raw rather than a fitted target. Rebuilding the target directly from the inverted quotes, by linear interpolation between the two expiries bracketing thirty days with no factor model at all, does not bear out the worry that we forecast an artificially smooth object; the direction is the opposite of the one usually assumed, since the raw series is slightly *more* persistent than the fitted one. The ridge's skill falls from 6.7% to 5.4% at fourteen days and from 13.9% to 13.2% at thirty, so smoothing accounts for a twentieth to a fifth of the measured skill depending on the horizon.

Length of the training window. Six years of training before the out-of-sample window opens might be too few, in which case the models would be handicapped rather than the series unforecastable. Moving the split date settles it. Training is expanding, so the figure quoted in the text is a burn-in and not a ceiling: by the end of the sample every model has been refitted on thirteen years. Across splits at January 2014, 2015, 2016, 2017 and 2018, which give 3.4 to 7.4 years of burn-in, the monthly skill of the ridge is 14.1, 14.2, 13.8, 13.8 and 14.6%. It does not move. What moves is the Clark–West statistic, from 2.14 down to 1.84 as the burn-in lengthens, and it moves in the direction that shows what the binding constraint is: a later split leaves a shorter *test* window, and it is the number of independent out-of-sample observations, about sixty-five at the monthly horizon, that limits the inference. More training data would not help; more years of options would.

Performance by volatility regime. Classifying days by whether IV_{30} exceeds its own expanding third quartile, a real-time rule, the strategy at $h = 30$ earns 1.22 in calm markets against a placebo at 0.81, and -0.67 in volatile ones against $+0.15$. Conditioning on regime does not rescue the forecast; it locates its failure in the crisis periods, which is where a volatility forecast would be worth most.

B Robustness and implementation

Curve construction. The alternative construction, which reads the vendor's implied volatilities at the listed strike nearest spot instead of inverting mid-quotes against a parity-implied forward, correlates 0.997 with the one used in the text but differs by 0.45 volatility points in mean absolute value, which exceeds the 0.3-point trading threshold: at $h = 5$, 20% of signal days carry an edge smaller than the gap between the two constructions on that day, falling to 9% at thirty days as the median edge rises from 0.90 to 1.72 points. Replaying the exercise on it changes nothing material. The strategy earns 0.33, 0.35 and 0.36 against a placebo at 0.35, 0.35 and 0.37, so the paired difference is -0.01, -0.00 and -0.01 where the main construction gives -0.09, -0.14 and -0.17; the spot-referenced variant earns -0.37, -0.32 and -0.08 against -0.43, -0.37 and -0.15. Every pair sits well inside a standard error of 0.33, and the sign of the conclusion is the same on both. The alternative construction is marginally kinder to the model, which is one reason we do not lead with it.

Penalty on the curve fit. The curvature carries most of what survives our multiplicity correction and is also the factor the daily ridge shrinks most, so its dependence on the penalty must be shown. At $h = 14$, removing the penalty entirely leaves the out-of-sample skill at 20.6% against 20.8% at our setting; raising it to $\alpha_{\mathrm{NS}} = 0.1$ gives 21.2%. Pushed to $\alpha_{\mathrm{NS}} = 1$, a penalty so severe that the standard deviation of β_2 collapses from 6.29 to 1.15 and the median fit degrades from $R^2 = 0.966$ to 0.804, the skill still stands at 16.9%. The effect is not an artifact of the regularization.

Code. The pipeline is implemented in Python: chain parsing and the Nelson–Siegel fit, the Box–Jenkins toolkit, the forecasters and evaluation, the futures-referenced rule with its placebo and the Lo (2002) standard errors, and the calendar spread. The standard-error routine is validated against the closed form $\sqrt{(1 + \hat{s}^2/2)/T}$ on i.i.d. draws before use, and a single script recomputes every number in the paper from the cached curves. All forecasts are strictly walk-forward, with an embargo of h observations for models using explanatory variables so that every competitor shares the same information set. The replication material opens with an automated guard in which every forecaster is run on a pure random walk, where skill must be indistinguishable from zero, since a one-day misalignment would appear as a skill near $1 - \sqrt{2} \approx -41\%$; all forecasters pass it.

C Modelling the futures curve directly

The paper forecasts a spot at-the-money volatility and then projects it into futures units, and two translations separate the object forecast from the object traded: the variance-swap wedge of Section 4.2 and the roll of Section 4. That leaves a question the rest of the paper cannot answer. Is there no signal, or is there a signal that both translations destroy?

The way to find out is to remove the translations. VIX futures settlements alone define a term structure, from which we build constant-maturity prices at thirty, sixty, ninety and one hundred and twenty days and hence a level, a slope and a curvature, all in the units of the traded contract. No option chain enters, so the sample is the futures record itself: 3,147 days

from January 2013 to February 2026, with 2013–2017 for training and **2018–2026** out of sample, some five years longer than the option window. We run two exercises on it.

The first forecasts the thirty-day constant-maturity *price* from the curve, which is the exact analogue of Section 3 in traded units. Table 11 reports it. The monthly skill is 5.3 to 5.5%, against 13 to 14% on the option-implied curve, and it is negative at every horizon out to a week. The futures price is more persistent than the volatility it tracks, with a first-order autocorrelation of 0.980, which makes the random walk harder to beat and accounts for most of the gap. The Clark–West statistics reach 2.08 and 1.85 at thirty days with bootstrap p-values of 0.036 and 0.050, the latter exactly at the boundary, so the monthly result is at the same boundary as everywhere else in this paper. One entry deserves a word: at one day the ridge has skill -0.6% and $t = 2.68$. There is no contradiction. Clark–West tests equality of population mean squared errors after removing the noise the extra parameters contribute, so it can reject while the sample error is larger; what it cannot do is make a model with negative skill useful.

The second forecasts the *return* of the constant-maturity position, which is the only formulation in which the object forecast and the object traded are the same thing. Measured against the expanding historical mean in the sense of Campbell and Thompson (2008), the out-of-sample R^2 is **negative** at all three horizons, from -4.3 to -5.6%: the curve's shape predicts the position's return worse than its own running average does. Table 12 reports it together with the trading test, where the placebo can no longer be the random walk, since in futures units the random walk gives an identically zero edge and therefore no position at all. We use the carry instead, short whenever the curve is in contango, which is the Simon and Campasano (2014) trade written in futures units alone and a more demanding benchmark than the random walk. The comparison is therefore *not* nested and we do not present it as such. The rule earns 0.01 to 0.12 of Sharpe against the carry's -0.27 to -0.19 over this window, a difference of $+0.27$ to $+0.31$ with t-statistics near 0.7.

Three conclusions follow, and the first is the one that matters.

Removing both translations does not recover a signal. The absence documented in the body of the paper is a property of the problem and not an artefact of forecasting one object and trading another. That was the largest question left open, and it is now closed in the direction the rest of the evidence pointed.

The second is a caution about this appendix itself. It is a third pass over the same instrument, and the discipline we impose in Section 3.3 applies to us here as much as anywhere: had one of these cells come out significant, it would have had to be counted against a search space that now includes it. We report it because the question was worth answering, not because the answer is favourable.

The third is a check on our own method, and it is the one positive finding of this appendix. The paper argues in Section 6 that pairing buys precision only above a correlation of about one half, the paired standard error scaling with $\sqrt{2 - 2\rho}$. Here the two arms' positions correlate at 0.06 to 0.17, far below that threshold, and the prediction holds exactly: the paired standard error is 0.38 to 0.41, *larger* than the 0.30 to 0.35 of either arm alone, so the upper bound on the difference is a nearly uninformative $+1.0$ to $+1.1$ against the $+0.40$ of Section 5. A non-nested placebo, however economically natural, forfeits what the nested one provides. That is the clearest demonstration in the paper of why the nested null is the right benchmark, and it is obtained on a sample and an instrument the main results never touch.

Table 11: Forecasting the constant-maturity VIX futures price directly from the futures curve, out-of-sample 2018–2026. Skill is the percentage reduction in root mean squared error against the random walk; t is the Clark–West statistic at $\max(2h, 7)$ Newey–West lags, in bold when its block-bootstrap p-value falls below five percent. AR(1) is anchored on the expanding mean. Compare the monthly skill of 5.3 to 5.5% here with the 13 to 14% obtained in Table 2 on the option-implied curve.

	Ridge		HAR		AR(1) anchored	
h	skill	t	skill	t	skill	t
1	−0.6	**2.68**	−0.5	0.93	−0.5	0.92
2	−3.4	0.54	−1.0	0.69	−1.7	0.64
5	−4.2	0.13	0.0	1.14	−3.2	0.83
7	−3.3	0.41	0.1	1.29	−4.0	0.92
14	0.3	**2.03**	2.6	1.80	−3.3	**1.86**
30	5.3	**2.08**	5.5	1.85	0.6	**2.35**

Table 12: Forecasting and trading the *return* of the constant-maturity position, out-of-sample 2018–2026. R^2_{OS} is measured against the expanding historical mean in the sense of Campbell and Thompson (2008), so a negative figure means the forecast is worse than that mean, while "corr" is the correlation between the forecast and the realized return, which can stay positive where R^2_{OS} is not. "Carry" is the placebo: short whenever the curve is in contango, which is the Simon–Campasano (2014) trade expressed in futures units alone. Each arm's dead zone is set at the same quantile of its own |edge| distribution, estimated on the training window only. Brackets are HAC standard errors; ρ is the correlation between the two arms' positions and the last column the ninety-five percent upper bound on the difference.

h	R^2_{OS} (%)	corr	Model	Carry	Difference	t	ρ	upper
5	−4.4	0.035	0.01 [0.30]	−0.27 [0.34]	0.27 [0.38]	0.72	0.06	1.02
14	−5.6	0.076	0.02 [0.32]	−0.25 [0.35]	0.27 [0.38]	0.71	0.06	1.02
30	−4.3	0.169	0.12 [0.35]	−0.19 [0.35]	0.31 [0.41]	0.75	0.17	1.12